

Data Table with Missing Values

 Customer ID	 Age	 Gender	 City	 Income	 Education	 Job Type	 Purchase Amount	 Rating	 Churn
1	25	Male	Dhaka	45000	Bachelor	Full-time	1200	4	No
2	31	Female	Chittagong	52000	Master	Full-time	1800	5	No
3	29	Male	Dhaka		Bachelor	Part-time	900	3	Yes
4	42	Female	Sylhet	76000	Master	Business	2500	5	No
5	36	Male	Rajshahi	61000	Bachelor	Full-time	1600	4	No
6	23	Female	Khulna	39000	Bachelor	Student	700	3	Yes
7	47	Male	Dhaka	88000	Master	Business	3200	5	No
8	33	Female	Barishal		Bachelor	Full-time	1400	4	No
9	28	Male	Rangpur	47000	Bachelor	Part-time	1100	3	Yes
10	39	Female	Dhaka	69000	Master	Full-time	2100	4	No
11	24	Male	Cumilla	41000	Bachelor	Student	800	3	Yes
12	52	Female	Sylhet	95000	Master	Business	4100	5	No
13	35	Male	Khulna	58000	Bachelor	Full-time	1700	4	No
14	30	Female	Dhaka	50000	n/a	Full-time	1500	4	No
15	41	Male	Chittagong	73000	Master	Business	2600	5	No
16	27	Female	Rajshahi	46000	Bachelor	Part-time	1000	3	Yes
17	44	Male	Dhaka	81000	Master	Full-time	2800	5	No
18	32	Female	Rangpur	54000	Bachelor	Full-time		4	No
19	26	Male	Barishal	43000	Bachelor	Student	850	3	Yes
20	38	Female	Cumilla	66000	Master	Full-time	2000	4	No



Blank Cell = Missing Value



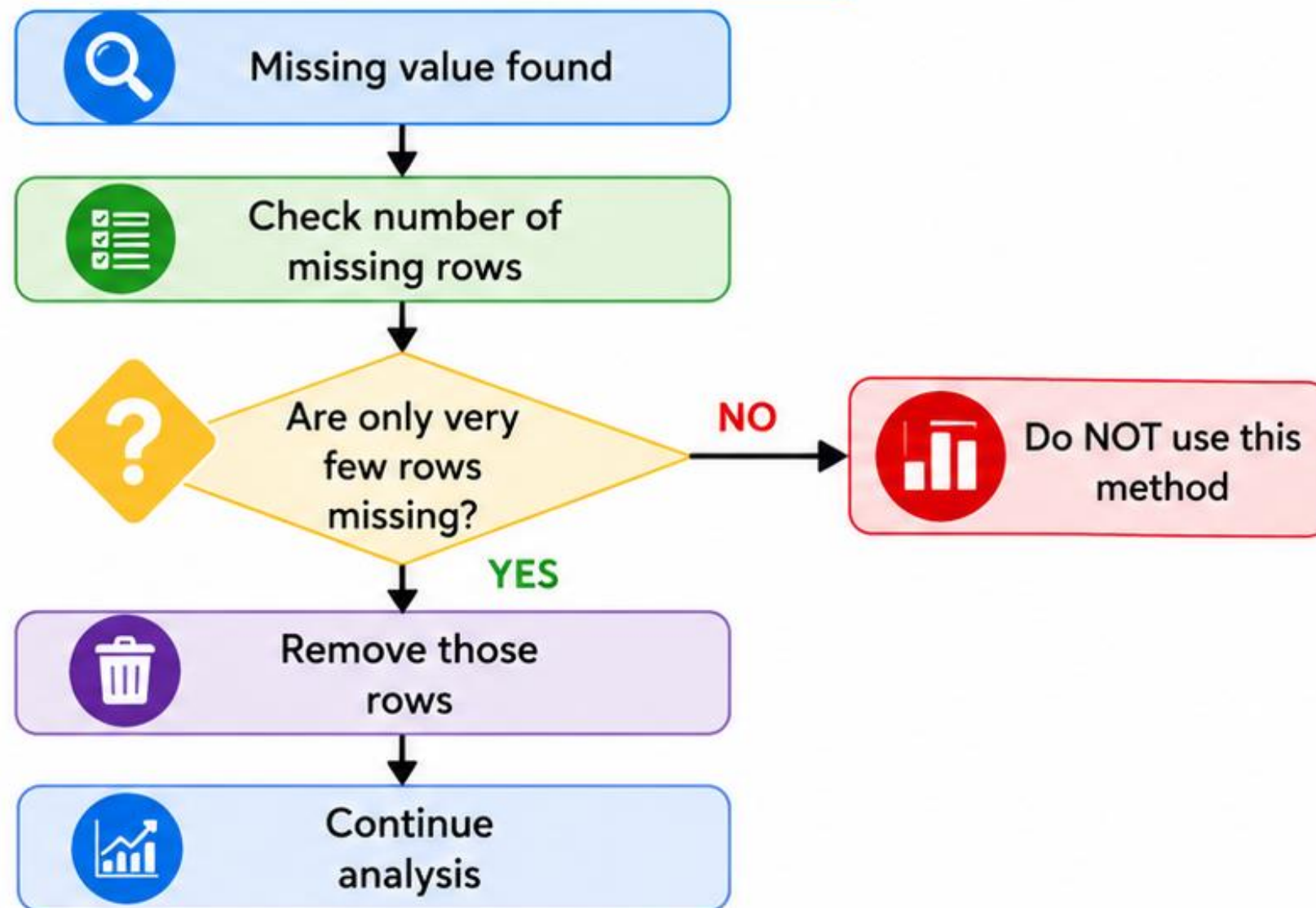
n/a = Missing Value



Method 1 – Removing Rows

This method means **deleting** the entire row where a missing value exists.

HOW IT WORKS (FLOWCHART)



USE THIS WHEN



Only a very small number of rows have missing values, and deleting them will not harm the dataset.



DO NOT USE THIS WHEN



Many rows have missing values, because removing them can reduce data size and create biased results.



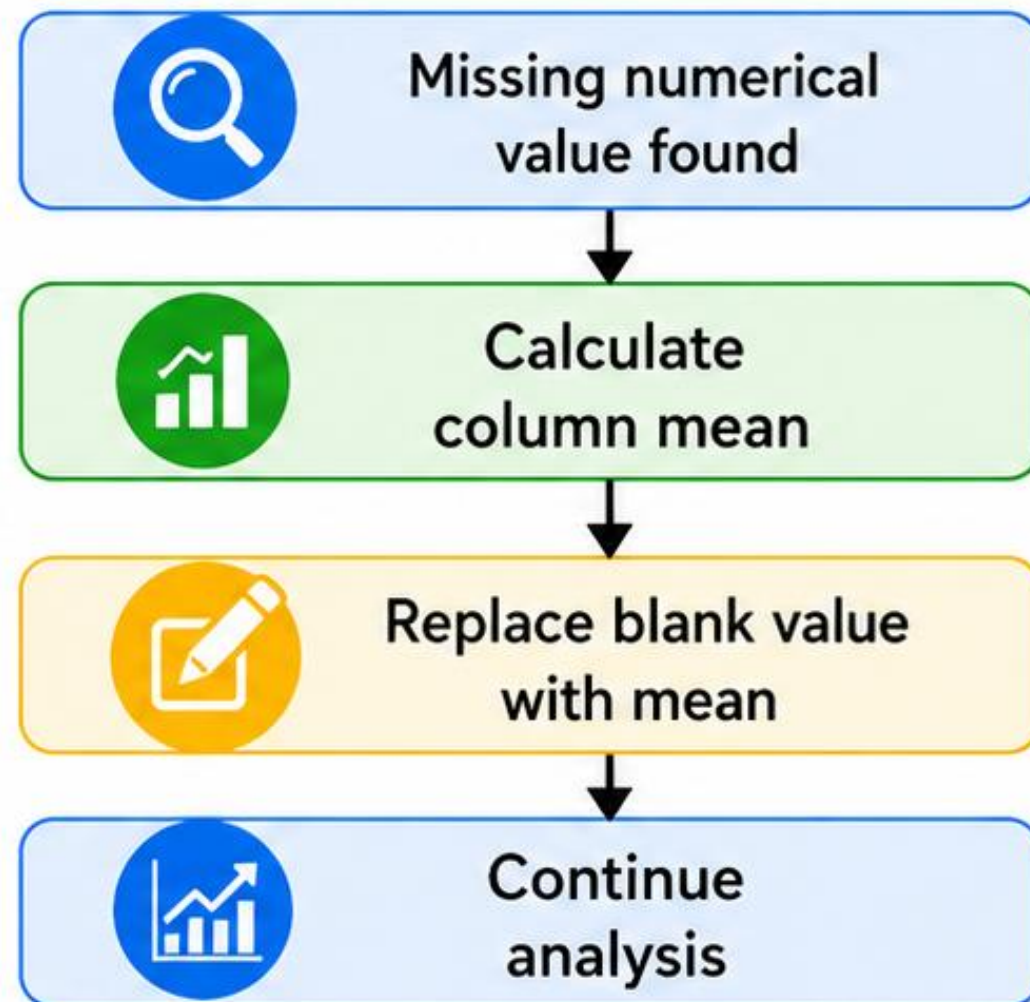
KEY TAKEAWAY: Use this method only when missing rows are very few and removing them will not affect the results.



Method 2 – Fill with Mean Value

This method replaces missing numerical values with the average value of that column.

HOW IT WORKS (FLOWCHART)



USE THIS WHEN



The numerical column has a fairly normal pattern and does not contain extreme values.



DO NOT USE THIS WHEN



The column has outliers, because the mean can be pulled too high or too low.



KEY TAKEAWAY: Mean imputation works well for numerical data with a normal pattern, but it is risky when outliers are present.



Method 3 – Fill with Median Value

This method **replaces** missing numerical values with the **middle value** of the column after sorting.

HOW IT WORKS (FLOWCHART)



USE THIS WHEN



The column has outliers, skewed data, or income-like values where a few very high or very low numbers can distort the average.



DO NOT USE THIS WHEN



The missing value depends on different groups and one overall median hides important group-level differences.



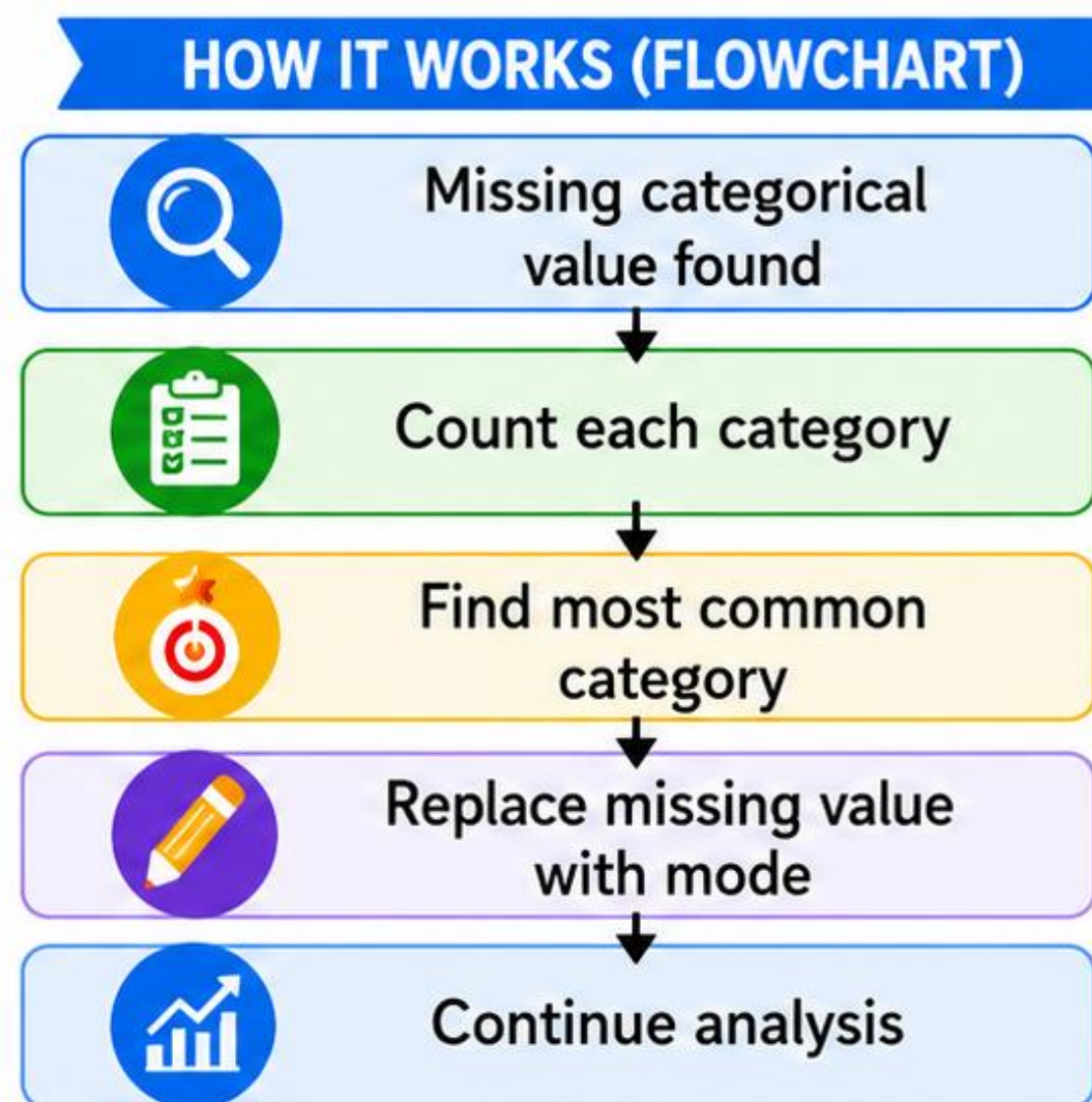
KEY TAKEAWAY: Median imputation is a strong choice for skewed numerical data and outliers, but one overall median may hide differences across groups.



Method 4 – Fill with Mode Value

This method replaces missing **categorical values** with the **most frequent category**.

HOW IT WORKS (FLOWCHART)



USE THIS WHEN



The column contains categories such as gender, city, education, or job type.



DO NOT USE THIS WHEN



The categories are highly balanced or the missing value has special meaning.



KEY TAKEAWAY: Mode imputation is useful for categorical data, but it is less reliable when categories are evenly distributed or when the missing value carries special meaning.



The BEST Method



Median imputation is often preferred because it is **more stable than the mean**. In real datasets, values like income, spending, age, delivery time, or transaction amount often contain extreme values. The **mean can be heavily affected** by these extreme values, but the **median stays closer to the typical central value**.



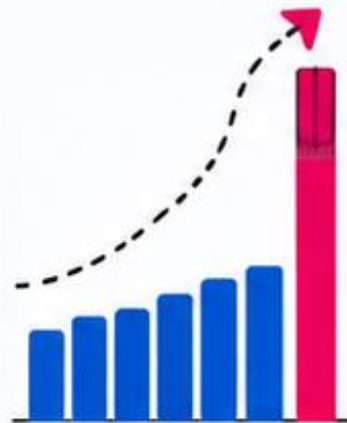
MEAN IS MISLEADING

Most customers earn between 40,000 and 70,000, but one customer earns **300,000**. The mean income becomes **misleading**.

MEAN

$$\frac{\text{Total Income}}{\text{Number of Customers}} = \mathbf{86,364}$$

(Pulled up by the outlier)



EXAMPLE: INCOME DATA (IN TAKA)

Customer	Income (Tk)
1	45,000
2	50,000
3	55,000
4	60,000
5	65,000
6	70,000
...	...
20	300,000

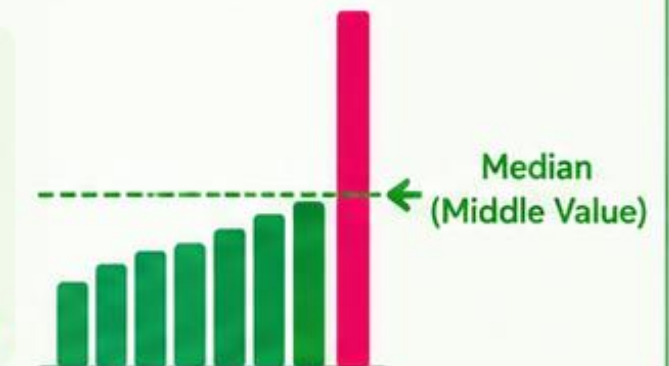
MEDIAN IS ROBUST



The median will still represent the **normal customer better**. That is why median imputation is usually **safer for skewed numerical data**.

MEDIAN

Middle Value
57,500
(Not affected by the outlier)



KEY TAKEAWAY

When data has outliers or is skewed, the **median** gives a **more reliable and representative value** compared to the mean.

